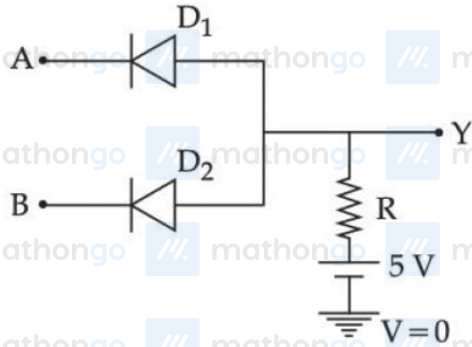


Q1 - 25 July - Shift 1

In the circuit, the logical value of $A = 1$ or $B = 1$ when potential at A or B is 5V and the logical value of $A = 0$ or $B = 0$ when potential at A or B is 0 V.

Space for your notes:



The truth table of the given circuit will be :

A	B	Y
0	0	0
(A) 1	0	0
0	1	0
1	1	1

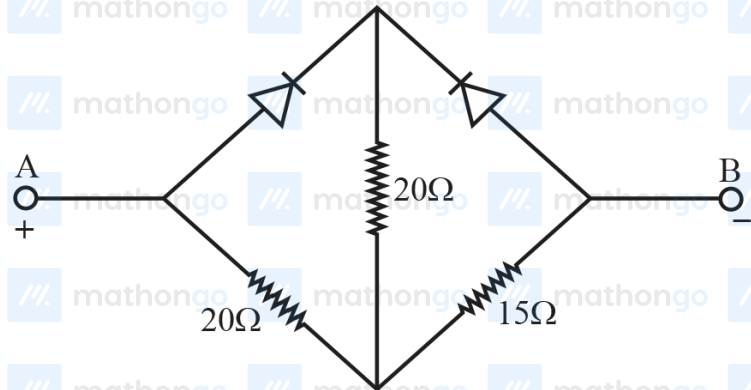
A	B	Y
0	0	0
(B) 1	0	1
0	1	1
1	1	1

A	B	Y
0	0	0
(C) 1	0	0
0	1	0
1	1	0

A	B	Y
0	0	1
(D) 1	0	1
0	1	1
1	1	0

Q2 - 25 July - Shift 2

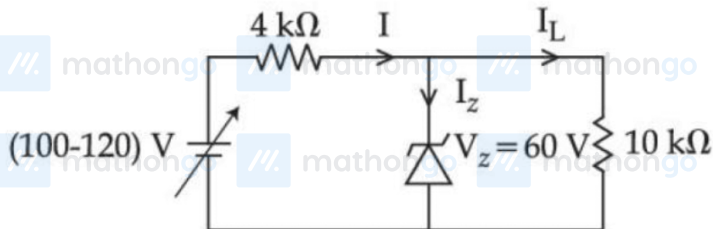
Two ideal diodes are connected in the network as shown in figure. The equivalent resistance between A and B is _____ Ω .



Space for your notes:

Q3 - 26 July - Shift 1

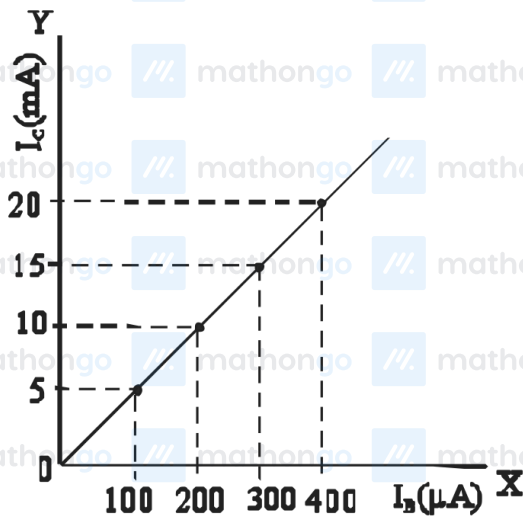
In the circuit shown below, maximum zener diode current will be _____ mA



Space for your notes:

Q4 - 26 July - Shift 2

The typical transfer characteristic of a transistor in CE configuration is shown in figure. A load resistor of $2\text{ k}\Omega$ is connected in the collector branch of the circuit used. The input resistance of the transistor is $0.50\text{ k}\Omega$. The voltage gain of the transistor is



Space for your notes:

Q5 - 27 July - Shift 1

Questions

MathonGo

A logic gate circuit has two inputs A and B and output Y. The voltage waveforms of A, B and Y are shown below



The logic gate circuit is

- (A) AND gate (B) OR gate
(C) NOR gate (D) NAND gate

Space for your notes:

Q6 - 27 July - Shift 2

For a constant collector-emitter voltage of 8V, the collector current of a transistor reached to the value of 6 mA from 4 mA, whereas base current changed from 20 μ A to 25 μ A value. If transistor is in active state, small signal current gain (current amplification factor) will be :

- (A) 240 (B) 400
(C) 0.0025 (D) 200

Space for your notes:

Q7 - 28 July - Shift 1

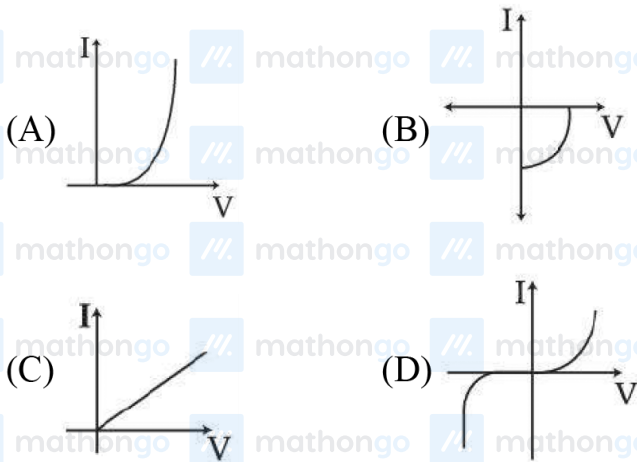
#MathBoleTohMathonGo

Questions

MathonGo

Identify the solar cell characteristics from the following options :

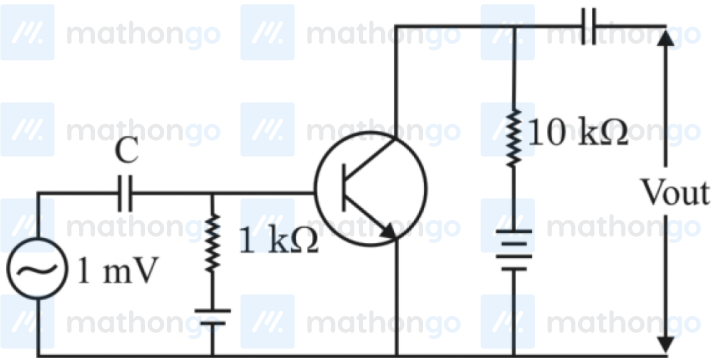
Space for your notes:



Q8 - 28 July - Shift 2

An n.p.n transistor with current gain $\beta = 100$ in common emitter configuration is shown in figure. The output voltage of the amplifier will be

Space for your notes:



- (A) 0.1 V (B) 1.0 V
(C) 10 V (D) 100 V

Q9 - 29 July - Shift 1

#MathBoleTohMathonGo

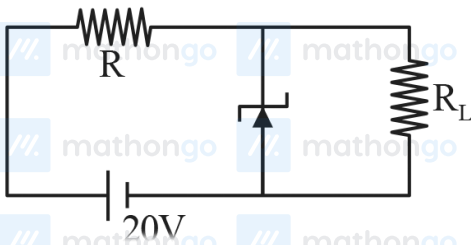
Questions

MathonGo

If the potential barrier across a p-n junction is 0.6 V. Then the electric field intensity, in the depletion region having the width of 6×10^{-6} m, will be _____ $\times 10^5$ N/C.

*Space for your notes:***Q10 - 29 July - Shift 2**

A 8 V Zener diode along with a series resistance R is connected across a 20 V supply (as shown in the figure). If the maximum Zener current is 25 mA, then the minimum value of R will be _____ Ω .

Space for your notes:

Answer Key

Q1 (A)

Q2 (25)

Q3 (9)

Q4 (200)

Q5 (A)

Q6 (B)

Q7 (B)

Q8 (B)

Q9 (1)

Q10 (480)

#MathBoleTohMathonGo

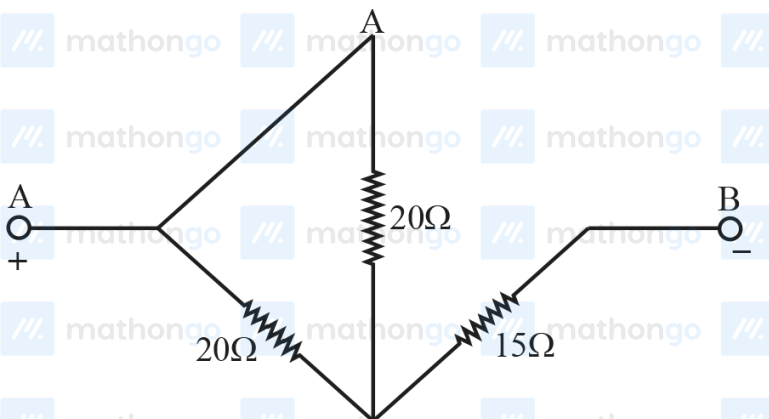
Q1 (A)

When both A and B have logical value '1' both diode are reverse bias and current will flow in resistor hence output will be 5 volt i.e. logical value '1'.

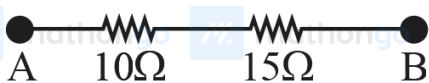
In all other case conduction will take place, hence output will be zero volt i.e. logical value '0'.

So truth table is

A	B	Y
0	0	0
0	1	0 (AND gate)
1	0	0
1	1	1

Q2 (25)

The forward biased diode will conduct while the reverse biased will not

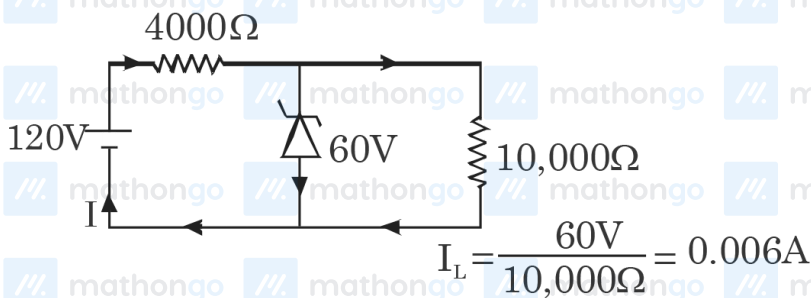


$$\therefore \text{Equivalent resistance} = 10 + 15 = 25\Omega$$

#MathBoleTohMathonGo

Q3 (9)

Consider input 120V



$$I = \frac{(120 - 60)V}{4000\Omega} = 0.015A$$

Thus $I_2 = I - I_L$

$$= 0.015 - 0.006 = 0.009 A = 9 \text{ mA}$$

Q4 (200)

Current gain in C-E configuration

$$\Rightarrow \beta = \frac{\Delta I_C}{\Delta I_B}$$

$$R_C = 2k\Omega, R_B = 0.50 k\Omega$$

$$\text{Voltage gain} = \frac{\Delta I_C R_C}{\Delta I_B R_B} = \frac{5 \times 10^{-3}}{100 \times 10^{-6}} \times \frac{2}{0.5}$$

$$= \frac{10^{-2}}{5 \times 10^{-5}} = \frac{1000}{5} = 200$$

Q5 (A)

#MathBoleTohMathonGo

By making Truth table

A	B	Output
0	0	0
1	1	1
0	1	0
1	0	0

Comparing with output of AND gate

A	B	AND
0	0	0
0	1	0
1	0	0
1	1	1

⇒ logic gate present is AND gate

Q6 (B)

#MathBoleTohMathonGo

$$V_{CE} = 8 \text{ V} = I_C = 6 \text{ mA from } 4 \text{ mA},$$

$$I_B = 20 \mu\text{A to } 25 \mu\text{A}$$

$$\text{Current gain } \beta_{av} = \frac{I_C' - I_C}{I_B' - I_B} = \frac{2 \text{ mA}}{5 \mu\text{A}}$$

$$\beta_{av} = \frac{2}{5} \times 10^3 = \frac{2000}{5} = 400$$

Q7 (B)

Conceptual / theory

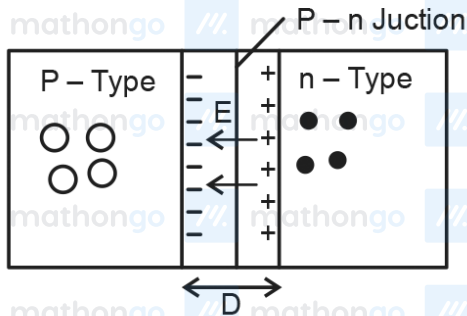
Q8 (B)

$$\frac{V_{out}}{V_{in}} = \beta \frac{R_{out}}{R_{in}}$$

$$V_{out} = \frac{100 \times 10 \times 10^3}{10^3} \times 10^{-3}$$

$$= 1 \text{ V}$$

Q9 (1)



$$E = \frac{V}{d} = \frac{\text{Potential barrier Across Junction}}{\text{width of Depletion layer}}$$

$$= \frac{0.6\text{V}}{6 \times 10^{-6}\text{m}} = 1 \times 10^5 \text{ V/m}$$

$$= 1 \times 10^5 \text{ N/C}$$

Q10 (480)

$$\varepsilon - IR - V_z = 0$$

$$20 - IR - 6 = 0$$

$$IR = 12$$

$$25 \times 10^{-3} R = 12$$

$$R = \frac{12}{25 \times 10^{-3}} = 480 \Omega$$

#MathBoleTohMathonGo